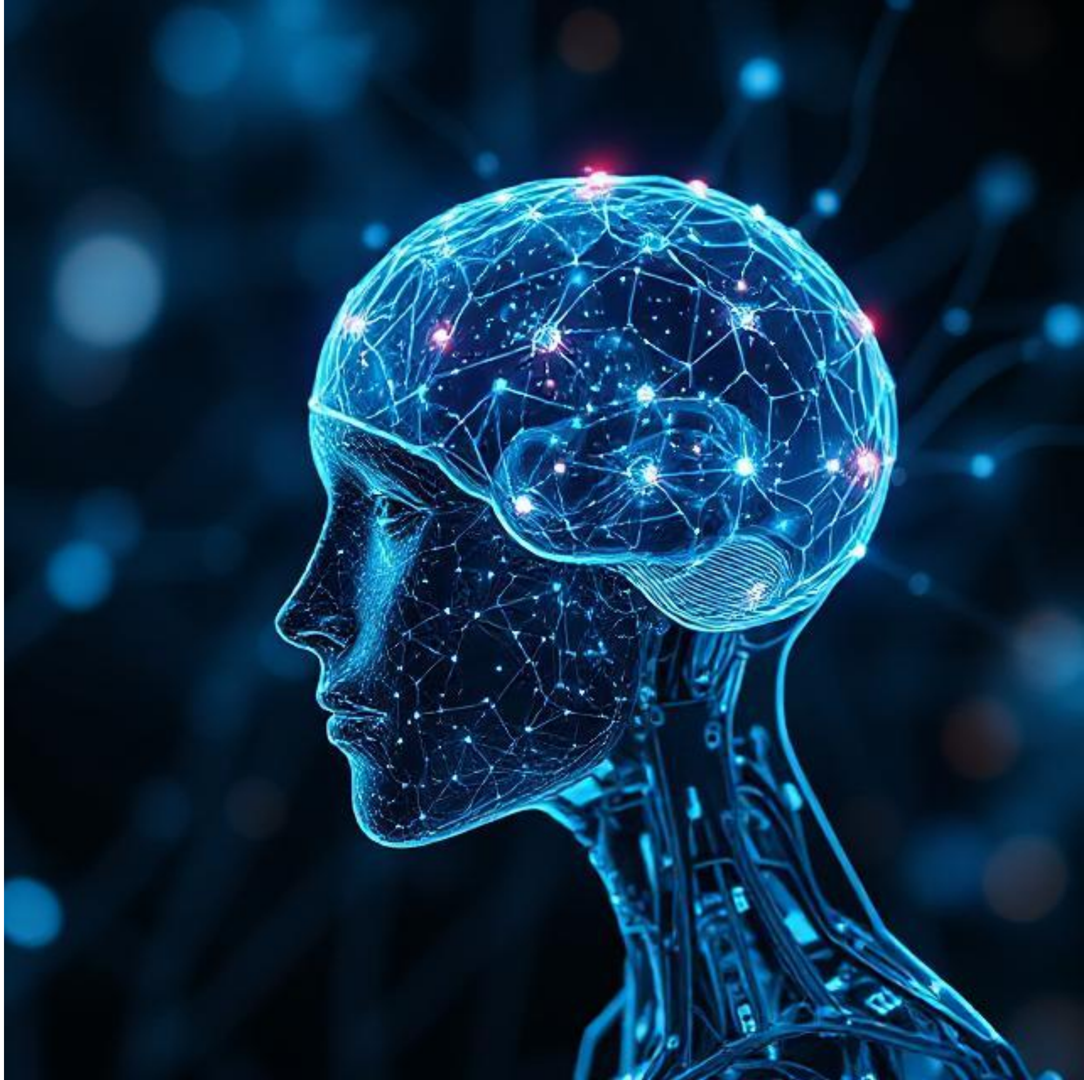
Deep Learning (38113)

Lecture 1: Introduction to Deep Learning



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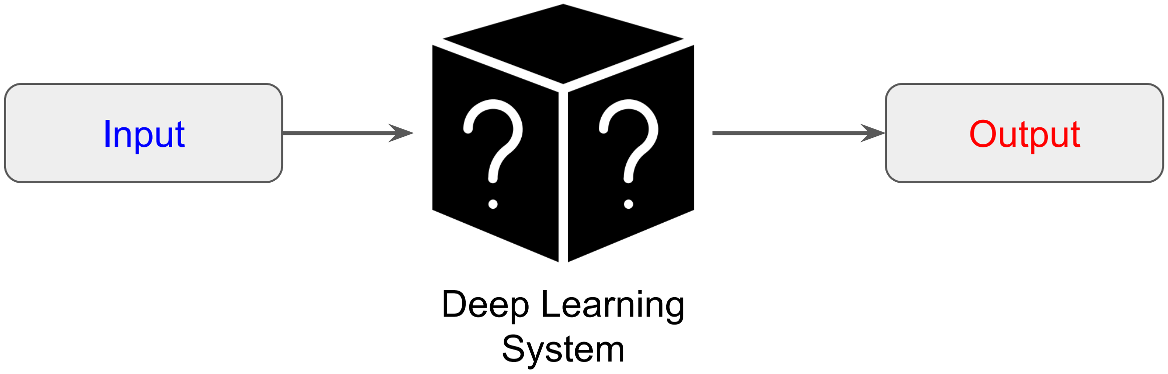
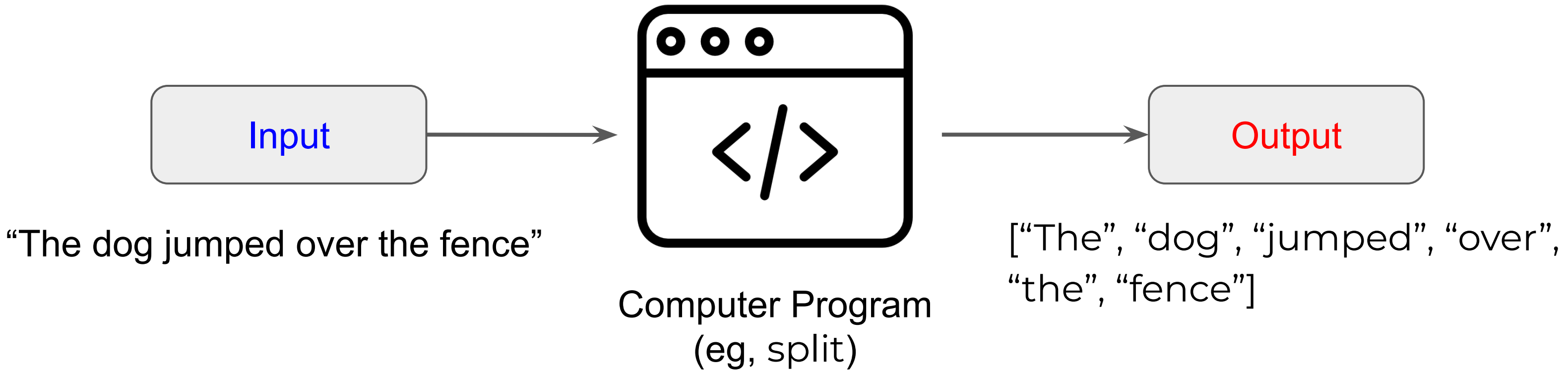
What is Deep Learning?



- A typical schema of a deep learning (DL) system:
    - Takes in an input
    - A “mystery box” that does some processing on the input
    - Produces an output

How is DL related to Programming?



- An example of computer program → split()
  - Input: "The dog jumped over the fence"
  - split(input, " ") → splits words separated by whitespace
  - Output is an array: ["The", "dog", "jumped", "over", "the", "fence"]

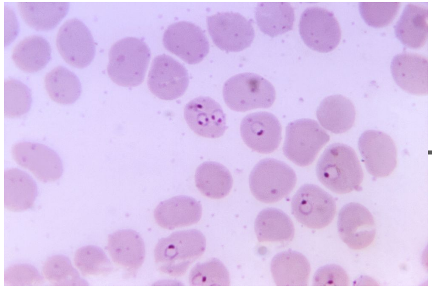
How is DL related to Programming?

```python
def simple_split(text):
    """Split text into words based on whitespace"""
    words = []
    current_word = ""

    for char in text: # iterate over each character
        if not char.isspace(): # if the character is not a whitespace
            current_word += char # store the character in current_word variable
        else:
            if current_word:  # Only add if we have a word
                words.append(current_word) # append the word to the array
                current_word = "" # make the current_word variable none

    # Add the last word if there is one
    if current_word:
        words.append(current_word)

    return words
```

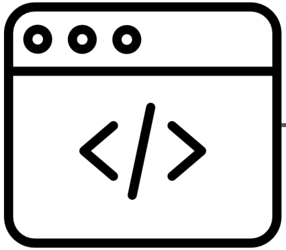
split() function in Python

- In computer programming we write explicit “rules” (or logic)
- The rules are applied to the input to get an output
- Other examples: sorting an array, binary search
- Pros:
    - Deterministic (or predictable) output
    - Interpretable
- Cons:
    - Works for simple input/output patterns

How is DL related to Programming?



Input: an image of red blood cells



Computer Program (count red blood cells)

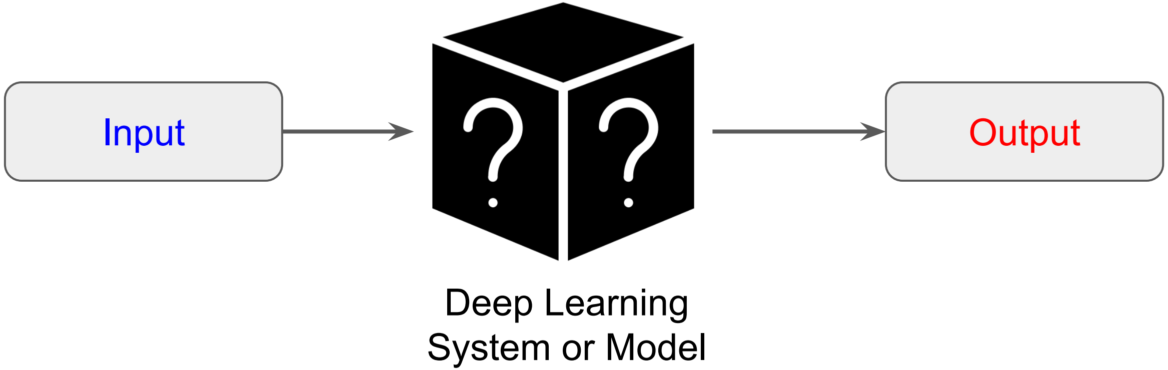
25

Output: number of red blood cells

- Question: How can we write “rules” for a complex problem such as “counting the number of red blood cells”?
- Answer: We can **not** trivially write a computer program that can count the number of red blood cells in an image ❌

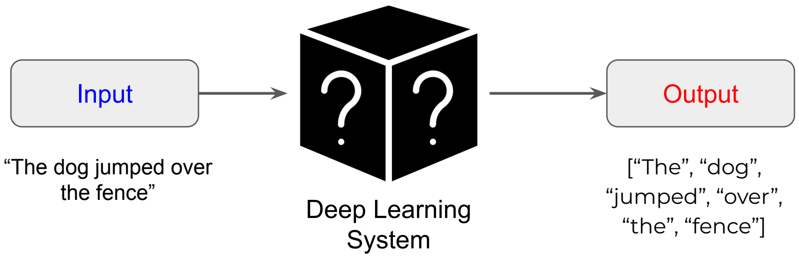
The image is licensed under CC0 1.0 Universal

DL to the rescue



- This is where DL shines → it can handle complex input/output patterns **without** the need to write complex rules by hand
- Ingredients in DL:
  - Example input/output pairs
  - A DL system or model

DL to the rescue



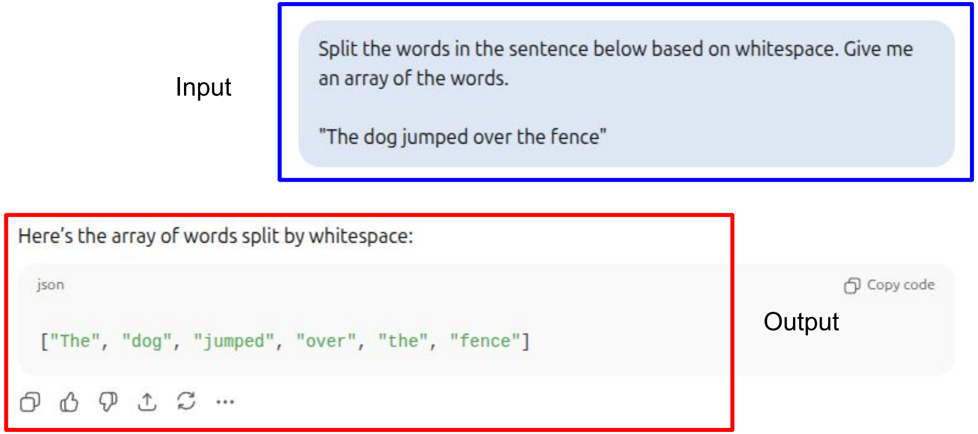
Example input/output pair

Input	Output
The cat scratched the sofa	[“The”, “cat”, …, “sofa”]
The kid spilled the glass of milk	[“The”, “kid”, …, “milk”]
…	…
Rome was not built in a day	[“Rome”, “was”, …, “day”]

- To make a DL system execute the task of splitting words by whitespace, we need to:
  a. Collect many examples of input/output pairs
  b. “Train” or “teach” the DL system (or model) the underlying rules
- Very similar to how we humans accomplish a task: we train ourselves!

DL to the rescue

Chat GPT
(a Deep Learning System)

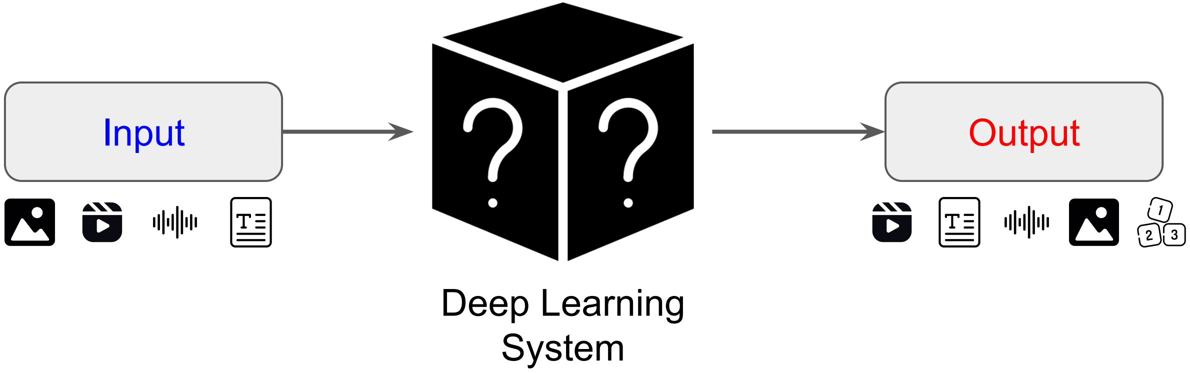


- The DL system Chat GPT was not explicitly programmed to split words based on whitespace
- It has “learned” from examples (or data) to produce a desired output given an input, a.k.a. the “rule”
- This is the power of Deep Learning! As long as you can get hold of examples you can teach it any “rules” you want (at least for most of the use cases)

Programming versus DL

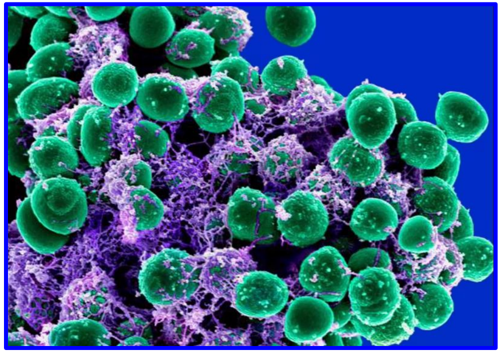
Computer Programming	Deep Learning (this course)
Human write explicit rules (eg, if-else, for loop)	Model learns rules from examples (or data)
Works well when the rules are clear and simple	Excels in complex tasks where writing rules is not feasible
Deterministic: predictable results	Probabilistic: learns patterns and makes predictions
Example: sorting numbers, binary search	Example: language translation (eg, EN → FR), count objects in an image (eg, count red blood cells)

Types of inputs and outputs in DL

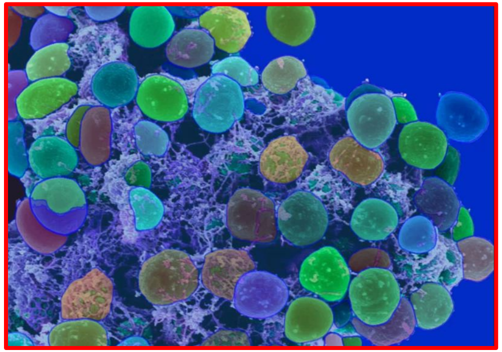


- The inputs and outputs to the model can be:
  - Text
  - Images
  - Audio
  - Combination of image + text
  - … anything you want

Examples of latest and greatest DL systems
(image → mask)



Segment Anything (from Meta)



Mask overlayed on input image

The images are taken from https://segment-anything.com/

Examples of latest and greatest DL systems (text → text)

“Welcome to the first ever "deep learning" course. In this course we are going to learn the inner working mechanisms of deep learning, and how to build cool AI applications using those techniques.”

Chat GPT

“Benvenuti al primo corso in assoluto di deep learning. In questo corso impareremo i meccanismi interni del deep learning e come realizzare applicazioni di intelligenza artificiale interessanti utilizzando queste tecniche.”

“Bienvenue dans le tout premier cours de « deep learning ». Dans ce cours, nous allons découvrir les mécanismes internes du deep learning et apprendre à créer de superbes applications d’intelligence artificielle en utilisant ces techniques.”

Examples of latest and greatest DL systems (text → image)

Prompt: “Generate a photo with the following theme: “Welcome to the first ever "deep learning" course. In this course we are going to learn the inner working mechanisms of deep learning, and how to build cool AI applications using those techniques.”

Gemini (from Google)

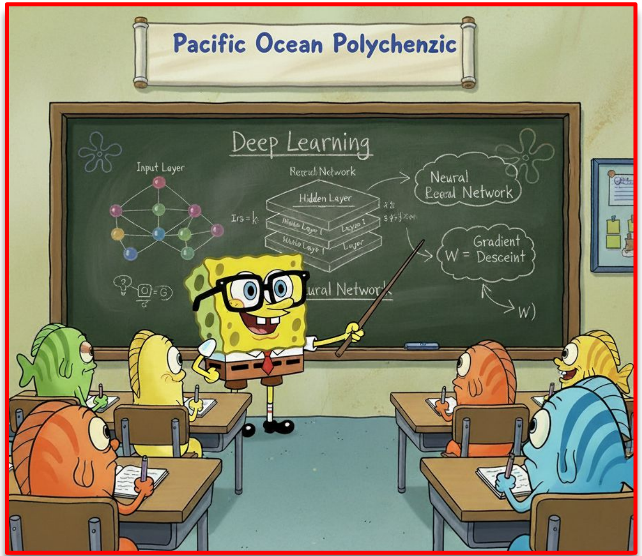
The image is generated using Google AI Studio.

Examples of latest and greatest DL systems
(text + image → image)

Text prompt: “Generate an image of the character below teaching deep learning”

Image: 

Nano Banana
(from Google)



The image is generated using Google AI Studio with Nano Banana.

Examples of latest and greatest DL systems (text → video)

Prompt: “In rural Ireland, circa 1860s, two women, their long, modest dresses of homespun fabric whipping gently in the strong coastal wind, walk with determined strides across a windswept cliff top. The ground is carpeted with hardy wildflowers in muted hues. They move steadily towards the precipitous edge, where the vast, turbulent grey-green ocean roars and crashes against the sheer rock face far below, sending plumes of white spray into the air.”

→ Veo 3 (from Google) →

The image is a video frame from VEO 3. More videos at: https://deepmind.google/models/veo/

Examples of latest and greatest DL systems (text + arrow keys → video)

Text prompt: “Jetski during the festival lights”
Keyboard prompts: arrow keys

Genie 3 (from Google)

The image is a video frame generated by Genie 3. More videos at: https://deepmind.google/discover/blog/genie-3-a-new-frontier-for-world-models/

Examples of latest and greatest DL systems
(image + arrow keys → video)

Image prompt:

Keyboard prompts: arrow keys

Genie 3 (from Google)

The left image is a famous painting by Edward Hopper. The image on the right is a video frame generated by Genie 3. Video source: reddit

Examples of latest and greatest DL systems
(video + text instructions → actions)

Video feed (not what the robot sees):

Text prompt: pick up the yellow funnel

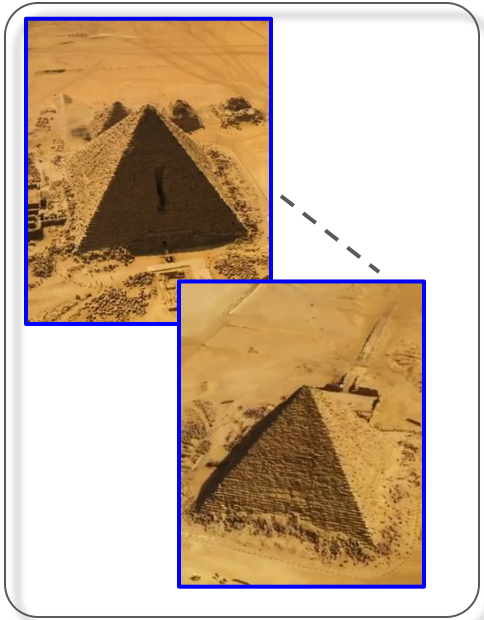
→ π →

pi-0.5 (from Physical Intelligence)

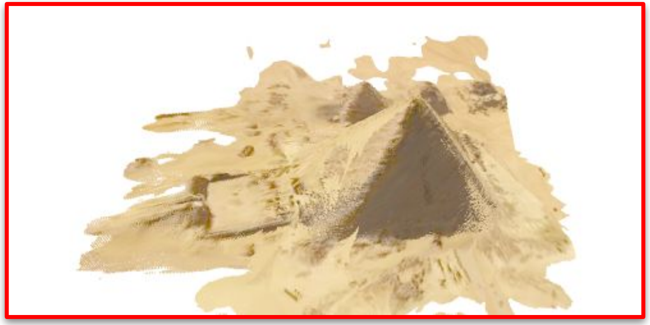
The robot performs an action described by the prompt

The image is a taken from Physically Intelligence Blog https://www.physicalintelligence.company/blog/pi05

Examples of latest and greatest DL systems
(image frames → 3D reconstruction)



VGGT (from Meta + Oxford University)

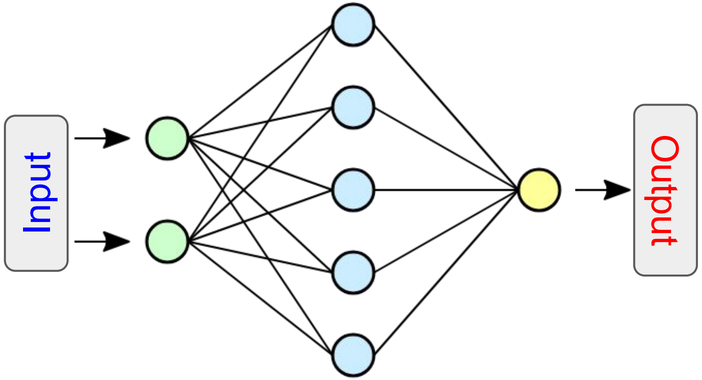


Check demo: https://huggingface.co/spaces/facebook/vggt

What’s inside a Deep Learning system?

VGGT (from Meta + Oxford University)

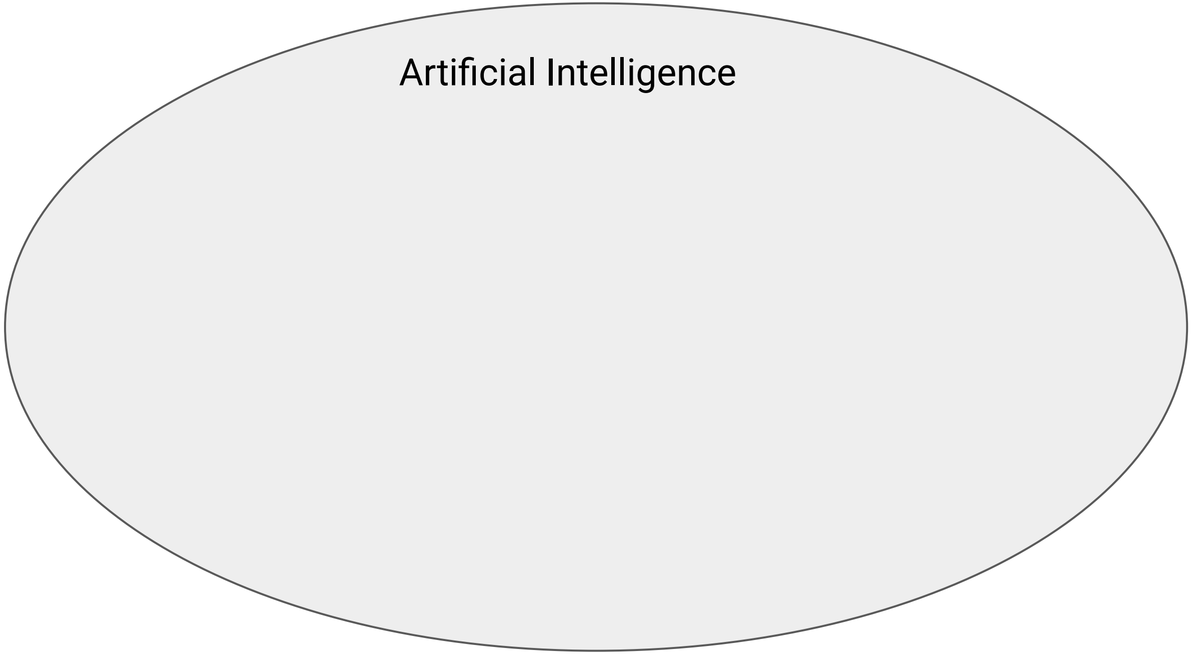
Segment Anything (from Meta)



A (simple) Deep Neural Network

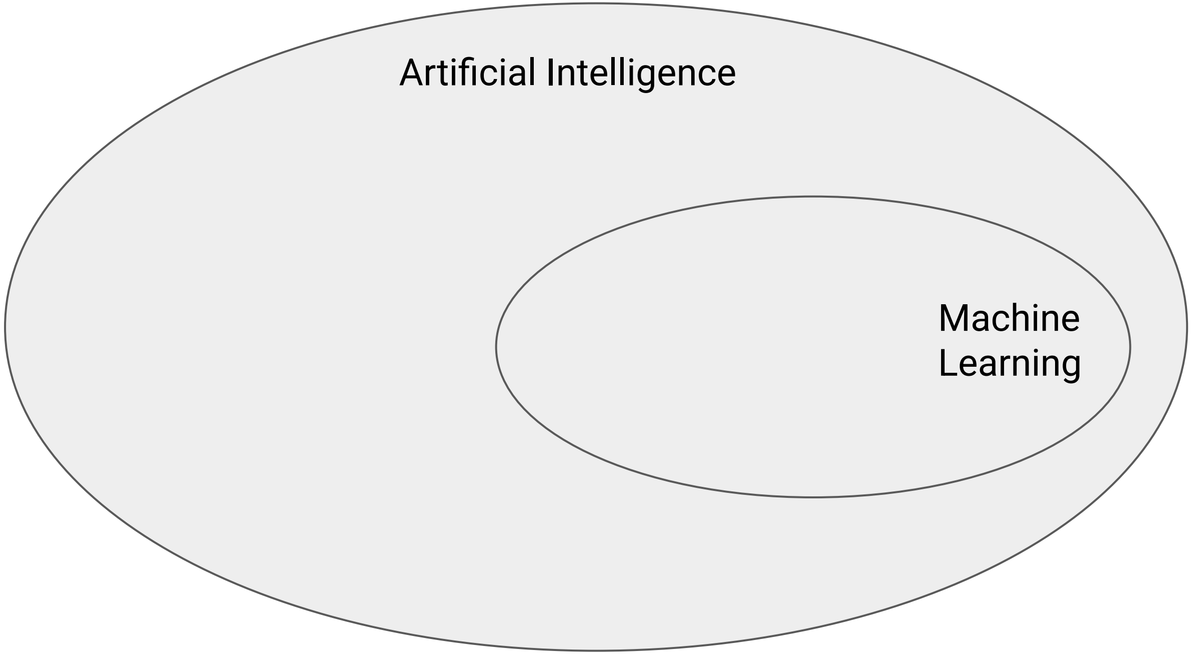
- Every deep learning system has one thing in common: a deep neural network
- Depending on the nature of input and output the “architecture” of the neural network will vary
- You will need to design the network based on the task you want to accomplish

First let's learn some jargons



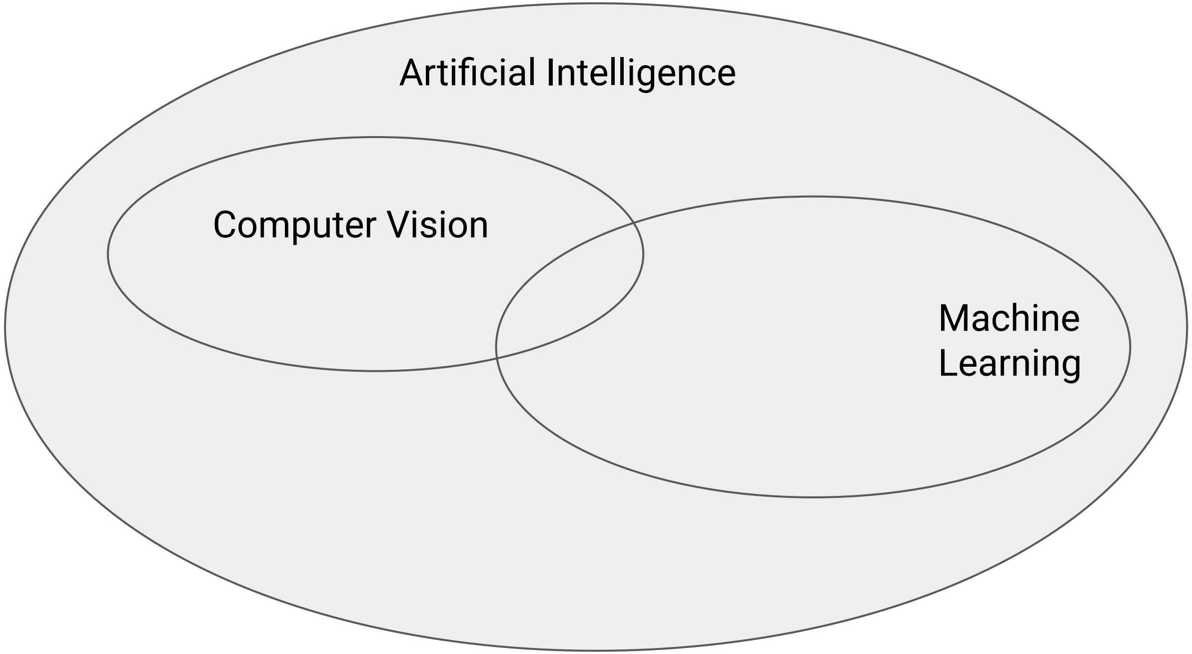
Artificial intelligence (AI) is the capability of computational systems to *perform tasks typically associated with human intelligence*, such as learning, reasoning, problem-solving, perception, and decision-making.

First let’s learn some jargons



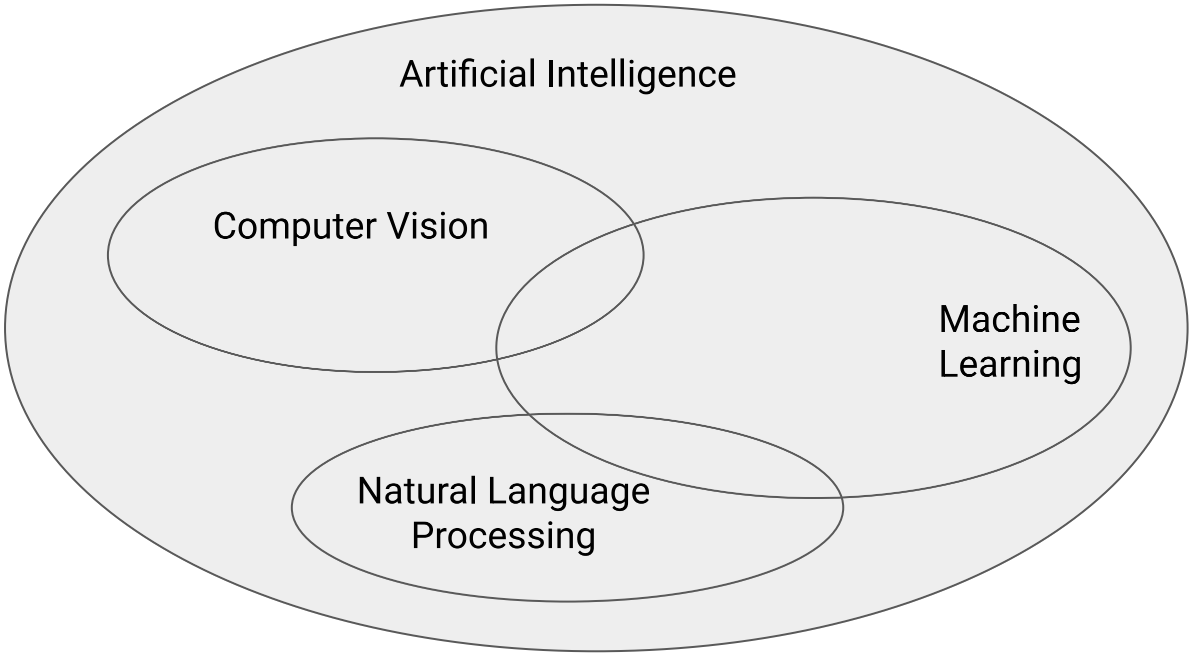
Machine learning (ML) is a field of study in artificial intelligence concerned with the development and study of statistical algorithms that can *learn from data* and generalise to unseen data, and thus *perform tasks without explicit instructions*.

First let's learn some jargons



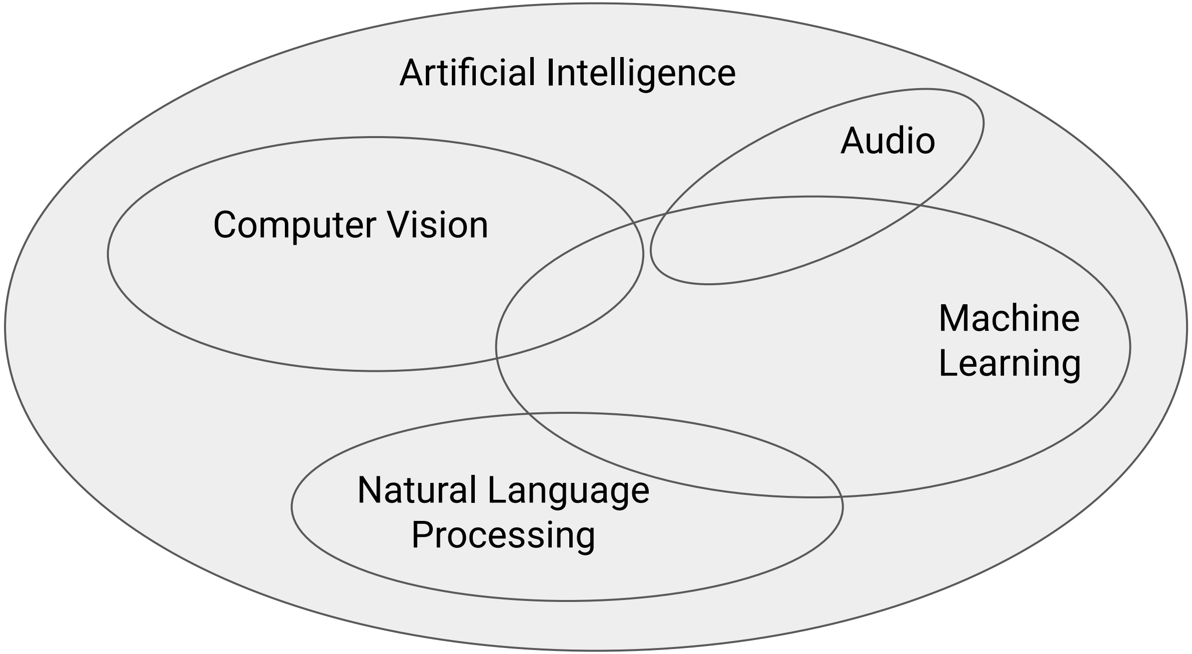
Computer vision tasks include methods for acquiring, processing, analyzing, and *understanding digital images*, and extraction of high-dimensional data from the real world *in order to produce* numerical or symbolic information, e.g. in the form of *decisions*.

First let's learn some jargons



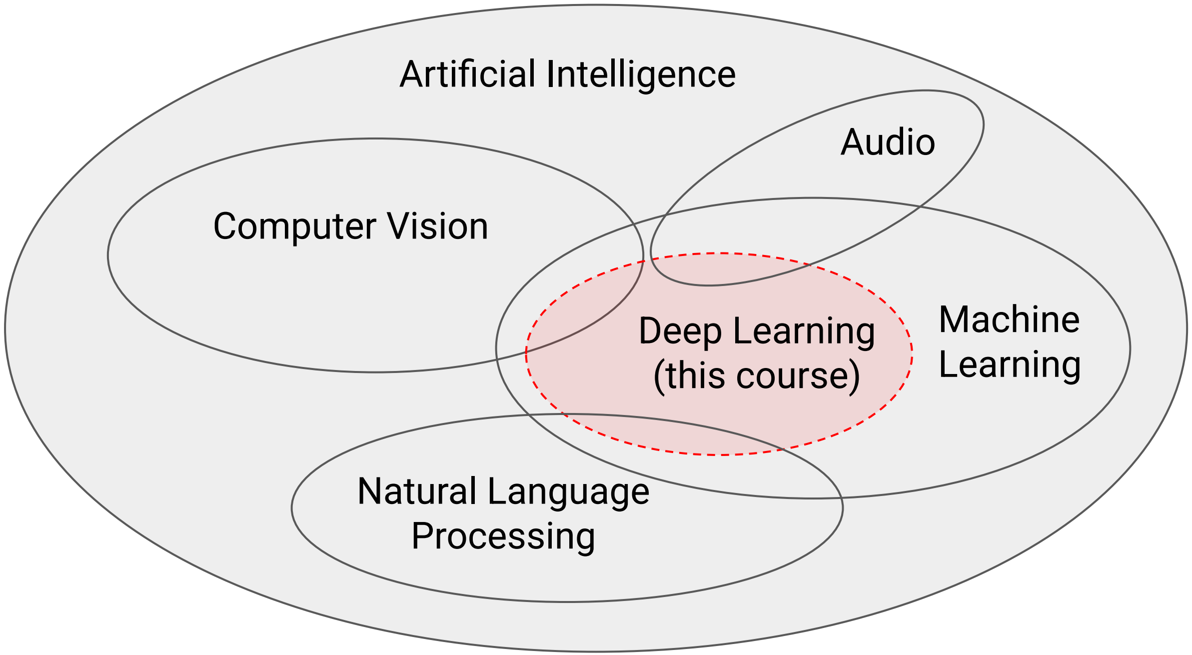
Natural language processing (NLP) is the *processing of natural language* information by a computer. The study of NLP, a subfield of computer science, is generally associated with artificial intelligence.

First let's learn some jargons



Audio signal processing is a subfield of signal processing that is concerned with the *electronic manipulation of audio signals*.

First let’s learn some jargons



Deep learning focuses on *utilizing multilayered neural networks* to perform tasks such as classification, regression, and representation learning.

What we will learn in this course?

- In this course we will explore the *foundations* and *applications* of modern **deep learning** systems.
- In particular we will explore:
  - The dark magic behind deep learning
  - Cutting-edge deep learning systems like Convolutional Neural Networks and Transformers
  - How to implement and train them ourselves
  - How can we apply these tools to power real-world applications

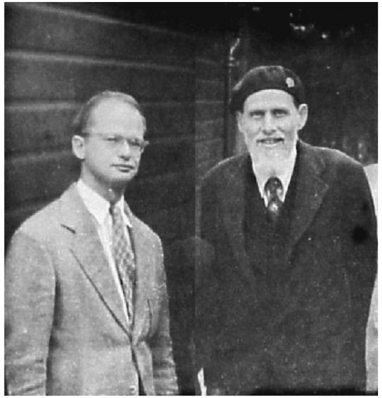
Lecture overview

- A brief history of deep learning breakthroughs
- Course overview
- Course logistics
- Demo

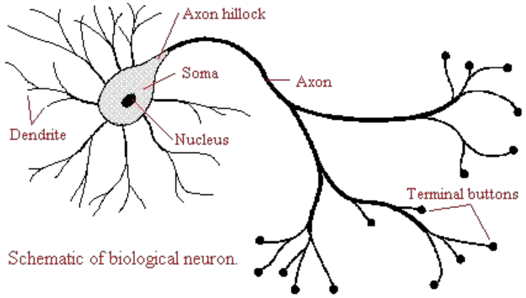
A brief history of Deep Learning Breakthroughs

McCulloch & Pitts: First artificial neuron model (1943)

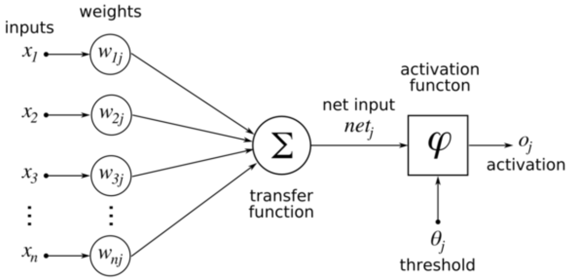
- The most fundamental unit of modern deep learning networks is an "artificial neuron"
- In 1943 McCulloch (neuroscientist) and Pitts (logician) laid the foundations by introducing the so-called "artificial neuron"
- The design of artificial neuron was *inspired* by biological neurons, which we can find in animal brains



Warren McCulloch and Walter Pitts



Biological neuron

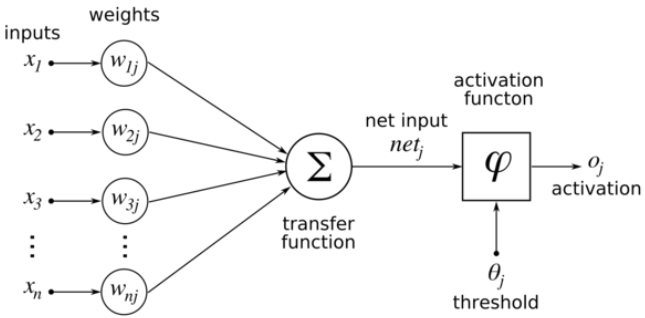


Artificial neuron

1943 McCulloch and Pitts

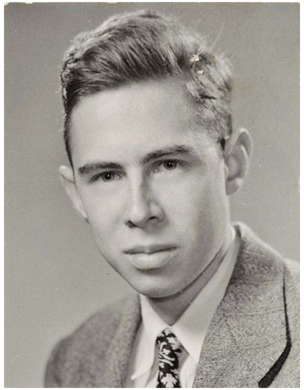
Rosenblatt's Perceptron (1957)

- The artificial neuron introduced by McCulloch and Pitts was more of a theoretical concept and there was no known algorithm to "train/learn" it (or adjust the values of w)
- Rosenblatt introduced the perceptron learning algorithm.
- What is perceptron? A perceptron *"is a binary classifier is a function that can decide whether or not an input, [...], belongs to some specific class"*

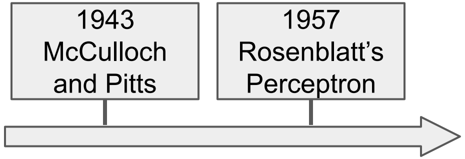


Given example input/output pairs, Perceptron learning algorithm finds the values of w.

Caveat: works for a simple class of problems (e.g. linearly separable data)

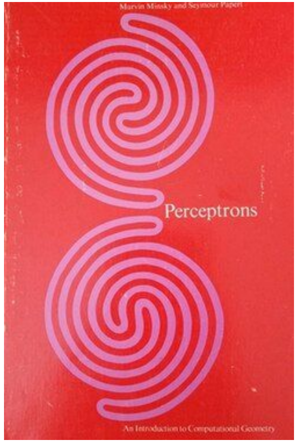


Frank Rosenblatt

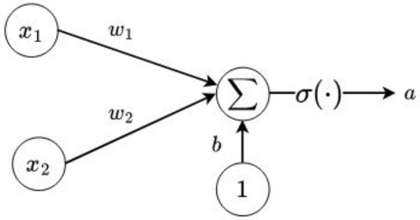


Minsky & Papert: Showed perceptron limitations (1969)

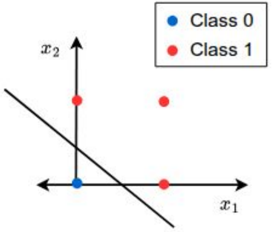
- The initial promise of Perceptron was hit hard by the publication by Minsky and Papert in 1969
- In the publication the authors imply that, “*since a single artificial neuron is incapable of implementing some functions such as the XOR logical function, larger networks also have similar limitations, and therefore should be dropped.*”
- This led to the first **AI Winter**



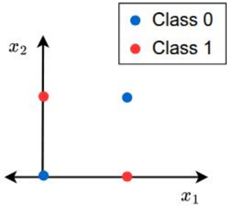
Perceptrons, 1969 by Minksy and Papert



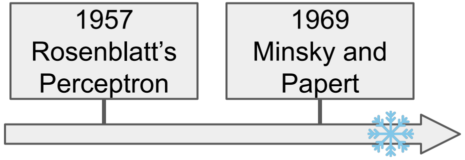
Perceptron with 2 inputs



OR logical function

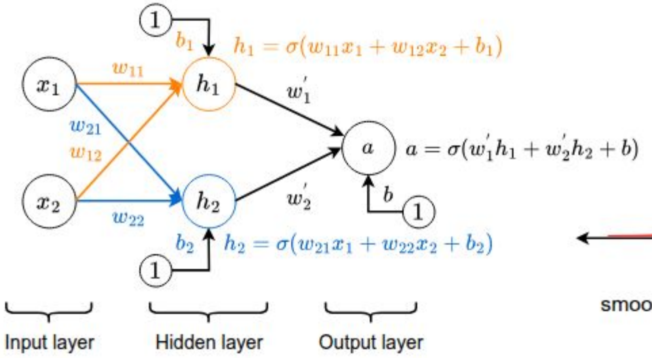


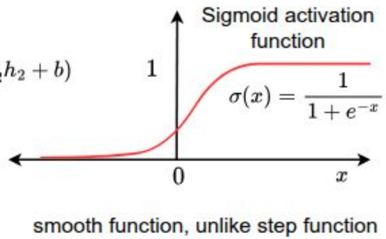
XOR logical function

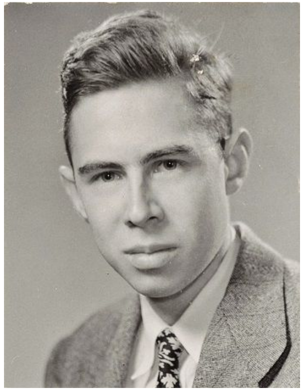


Small detour: Multilayer Perceptron (1958)

- In fact the XOR problem can in theory be solved by using a Multilayer Perceptron (MLP) and was invented by Rosenblatt in 1958
- What is MLP? It's a **network** of many Perceptrons and has at least one "hidden layer". That's why it's also called "**deep learning**" (coined later in 2006)





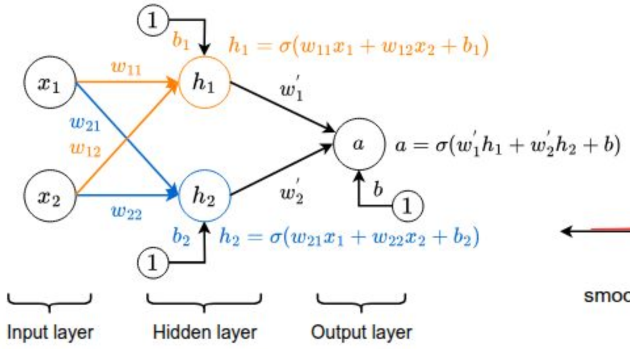


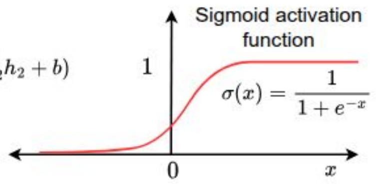
Frank Rosenblatt

- MLP could solve the XOR problem but there was no way to train/learn the weights w.

Rumelhart, Hinton & Williams: Backpropagation (1986)

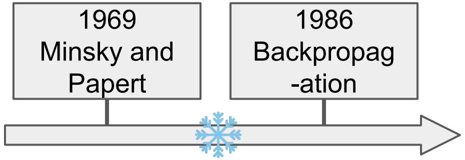
- It took another 20 years by Rumelhart, Hinton and Williams to invent one of the most influential algorithms → **Backpropagation**
- What is backpropagation? It’s an algorithm to *efficiently* train MLPs (no matter how deep it is)





smooth function, unlike step function

Geoffrey Hinton

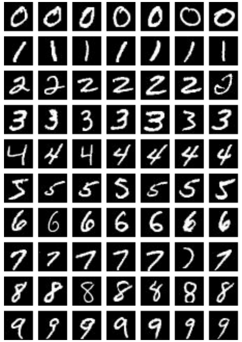


- It is still used today by every modern deep learning systems.

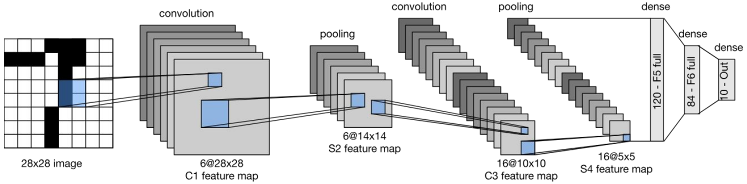
Yann LeCun: LeNet-5 (1998)

- The first success in recognizing images (ie, image classification) happened through the introduction of LeNet-5 by Yann Lecun
- What is LeNet-5? It’s a Convolutional Neural Network (CNN) that takes an image as input and outputs a class label
- It is very similar to modern day CNNs
- It was trained using MNIST dataset, which is a dataset of handwritten digits

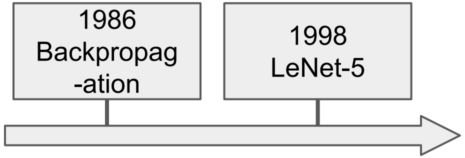
Yann Lecun



MNIST digits



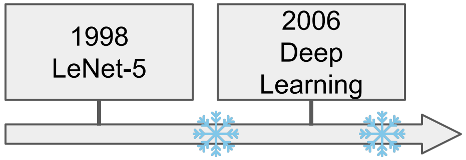
LeNet-5



Watch the video of live demonstration by Yann Lecun: https://youtu.be/FwFduRA_L6Q?si=TZh1Frt0VJL-E1Ay

Deep Learning in the 2000s

- Progress in “deep learning” was slow in the 2000s because:
  - It was compute intensive to train very deep neural networks
  - There existed another family of computational models called Support Vector Machines (or SVM) that were mathematically rigorous and easy to train
- Notable works on “deep learning” in the 2000s:
  - Restricted Boltzmann Machines
  - Deep Belief Networks
- Often considered as the second **AI Winter**

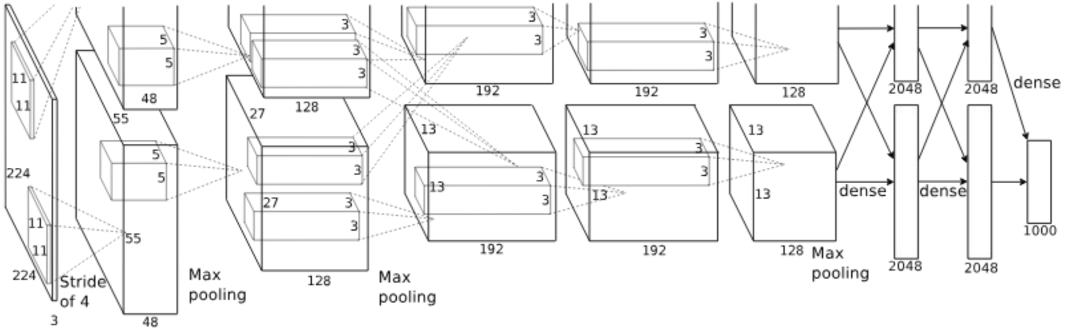


AlexNet wins ImageNet Challenge (2012)

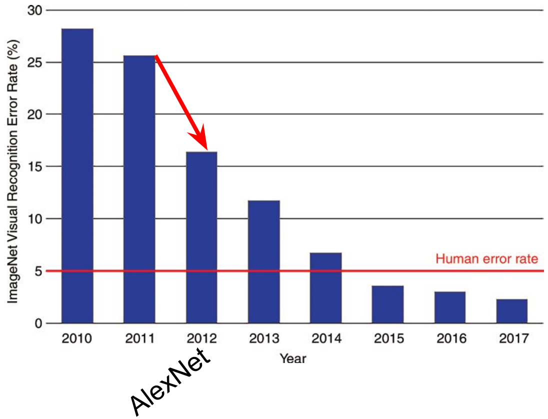
Image Classification Challenge
1000 object classes
~ 1.2 million images

Example images from ImageNet

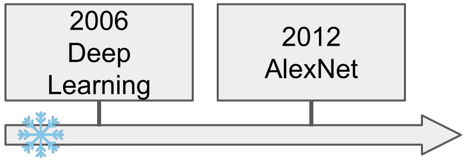
AlexNet wins ImageNet Challenge (2012)



AlexNet (a deeper CNN)

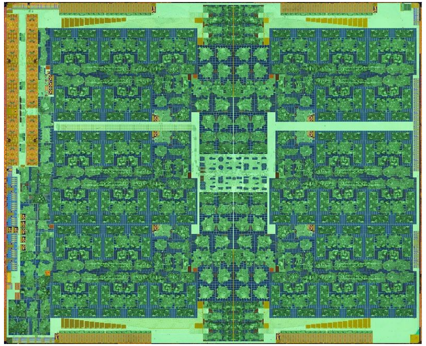


- AlexNet was a milestone work because:
  - For the first time drastically reduced the error rate in ImageNet challenge
  - Trained a deep neural network using GPUs



Detour: What are GPUs?

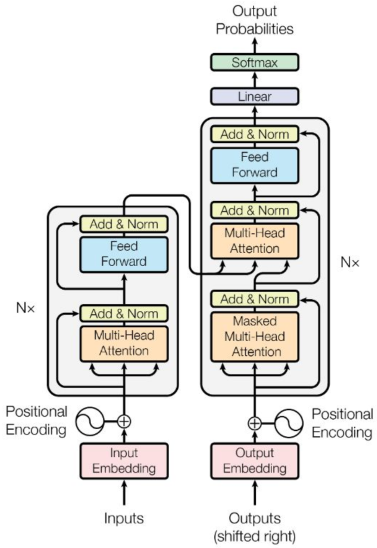
- A GPU or **graphics processing unit** is a specialized electronic circuit designed for digital image processing and to accelerate computer graphics
- Extensively used for playing video games
- Quintessential for serious deep learning
- Why GPUs are so useful?
    - GPUs are built for super fast *matrix multiplication*
    - Video games need super fast rendering of scenes (which is nothing but tons of matrix multiplication)
    - In deep learning we do nothing but mostly matrix multiplication
- Examples: RTX 3090 (consumer grade GPUs), A100 (server-grade GPUs)

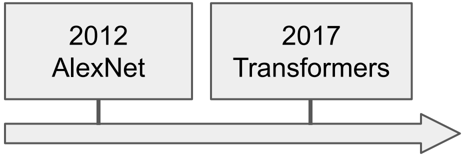


Inside a GPU

Transformers: Attention is all you need (2017)

- A major breakthrough in deep learning happened in 2017 with the introduction of Transformers
- A transformer is “*network architecture based solely on attention mechanisms, dispensing with recurrence and convolutions entirely.*”
- Forms the backbone of every modern deep learning systems such as Chat GPT and Claude
- Why is it used extensively?
  - flexible architecture, can be used for any kind of data such as images, text, and videos
  - can be scaled up
  - highly parallelized computation or GPU-friendly





The modern era (2018-Present)

- Many follow-up works have shown how Transformers can be used to address a plethora of tasks

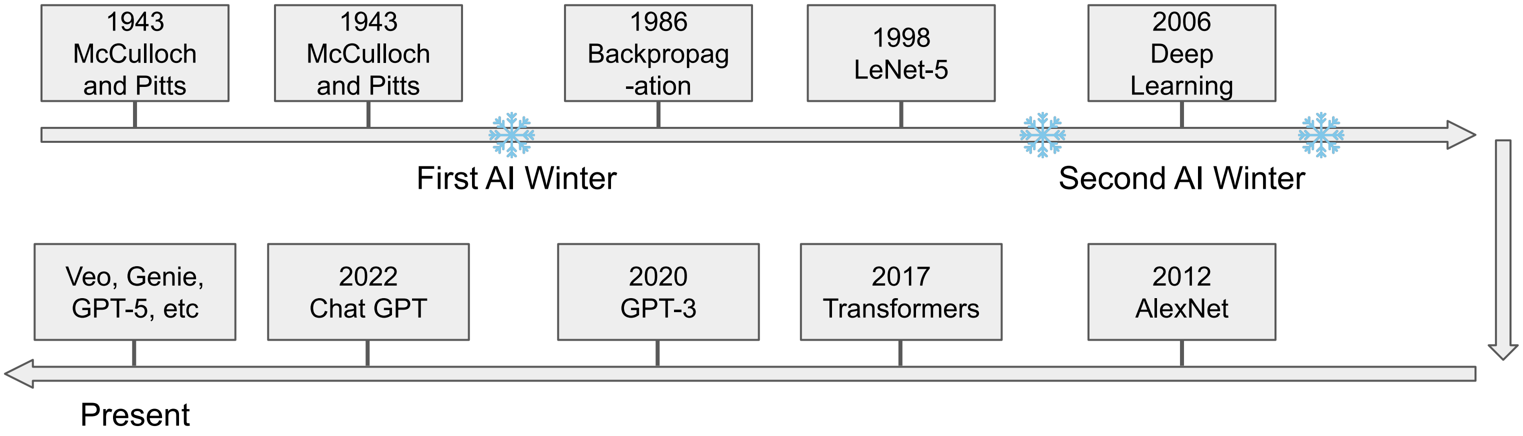
Natural Language Processing
- text summarization
- coding
- question answering
- …

Computer Vision
- image classification
- segmentation
- image matching
- …

Generative modelling
- text–to–image/video generation
- text–to–game generation
- image captioning
- …

Audio understanding
- automatic speech recognition
- audio generation
- text–to–speech generation
- …

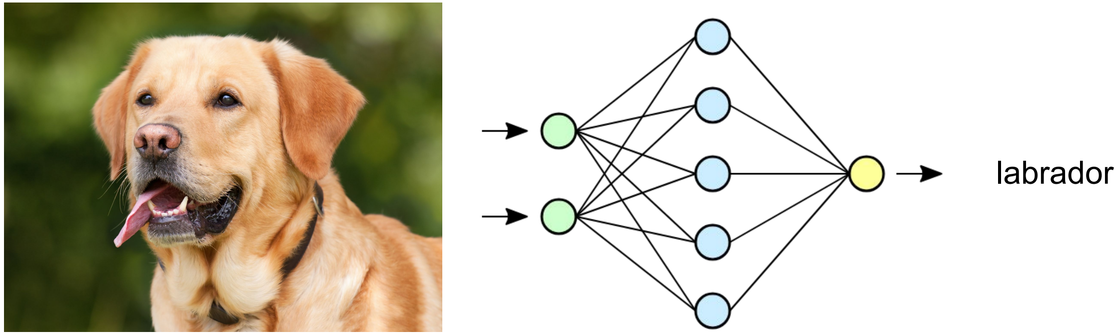
A brief history of Deep Learning Breakthroughs



Course overview

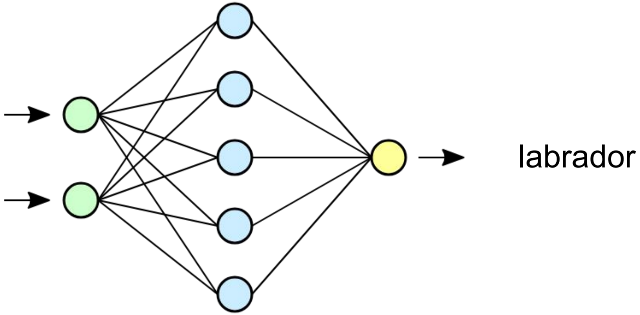
Fundamentals of Deep Learning

- We will learn the basics of deep learning using a core task in computer vision → image classification
- In image classification the goal is to predict the class of the object visible in the image



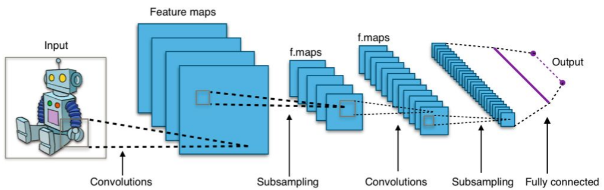
Fundamentals of Deep Learning

- In the context of image classification we will topics such as:
  - optimization
  - regularization
  - training deep neural networks
  - … many other essential concepts

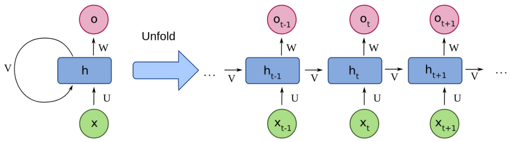


Advanced Deep Neural Networks

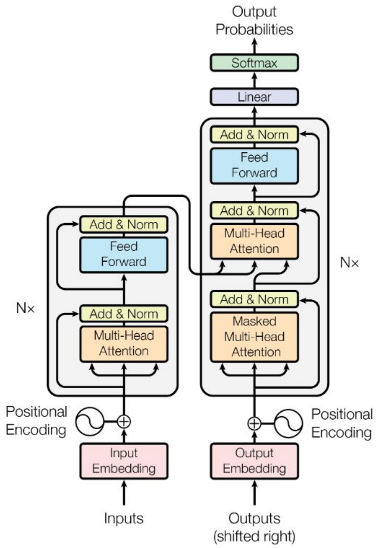
- We will go beyond "vanilla" deep neural networks and learn more widely used neural networks such as:
  - Convolutional Neural Network (CNN)
  - Recurrent Neural Network (RNN)
  - Transformers



CNN
(typically for images)



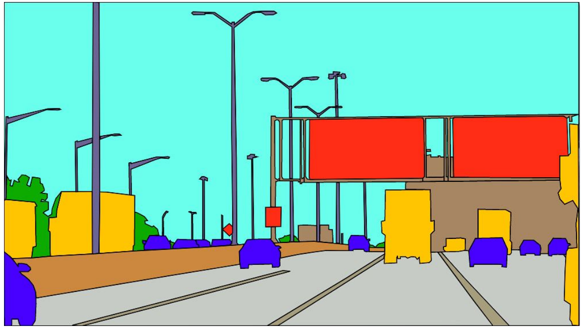
RNN
(typically for text)



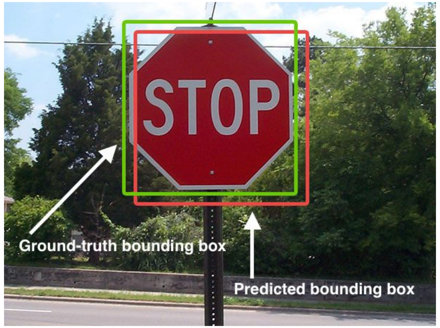
Transformer
(for almost every kind of data)

Beyond Image Classification

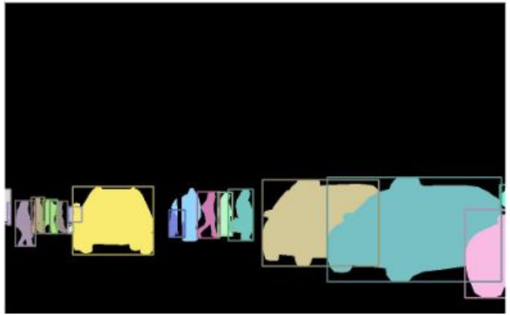
- We will learn about tasks that are more advanced that image classification:
    - Semantic segmentation
    - Object detection
    - Instance segmentation
- In all the above tasks we want to predict at a more fine-grained level than image classification
- Such tasks are essential in autonomous driving scenarios



Semantic segmentation



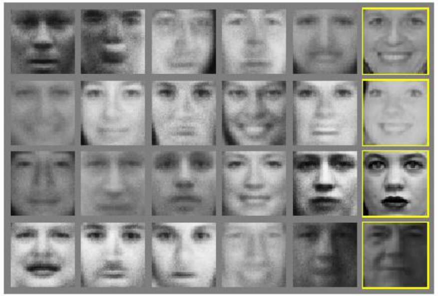
Object detection

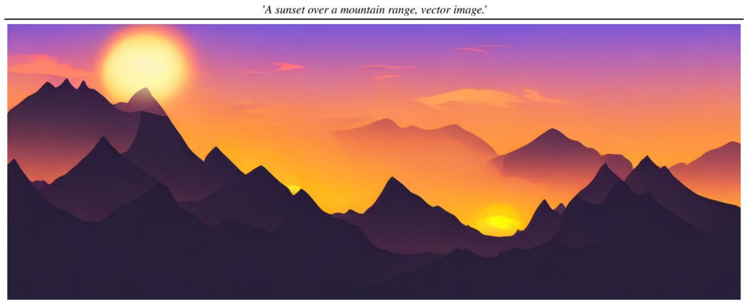


Instance segmentation

Basics of Generative Modelling

- We will learn the fundamentals of generative modelling, which for instance allows us to generate new images from noise or text
- In particular we will look at:
  - Autoencoder (AE)
  - Variational Autoencoder (VAE)
  - Generative Adversarial Network (GAN)
  - Diffusion Models





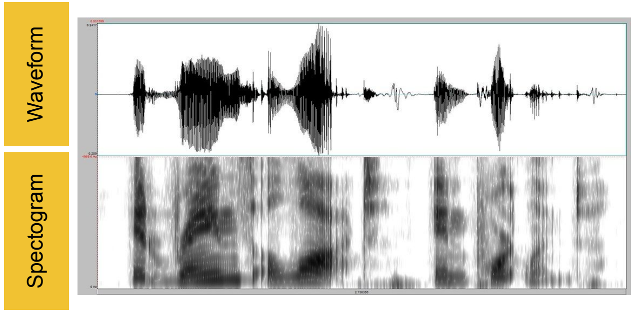
Natural Language Processing with Deep Learning

- We will learn about natural language processing tasks such as:
  - sentiment classification
  - text generation
  - language translation
  - instruction tuning

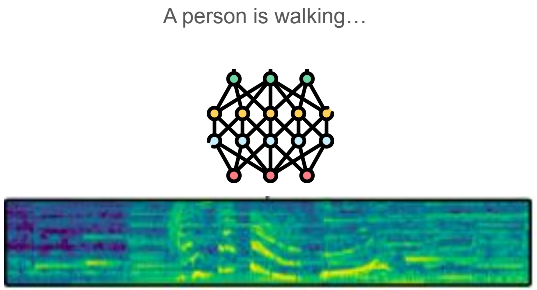
Input: “The movie was amazing!”	Input: “Once upon a time…”	Input: “Good morning” (English)	Input: “Summarize: Deep learning uses neural networks to learn patterns”
Output: Positive	Output: “there was a robot who loved to learn.”	Output: “Buenos días” (Spanish)	Output: “Deep learning = neural networks learning patterns.”
Sentiment classification	Text generation	Language translation	Instruction tuning

Audio Processing with Deep Learning

- In the domain of audio processing we will learn:
  - Spectrograms, waveforms → turning sound into visual-like data for DL
  - CNN-based audio classification
  - Automatic speech recognition (ASR), ie, map raw audio → text
  - Wav2Vec, which is a model for learning from raw waveforms
  - Whisper, a multilingual ASR



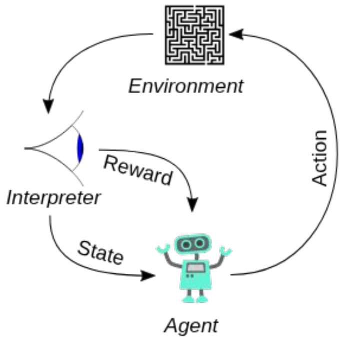
Waveform vs spectrogram

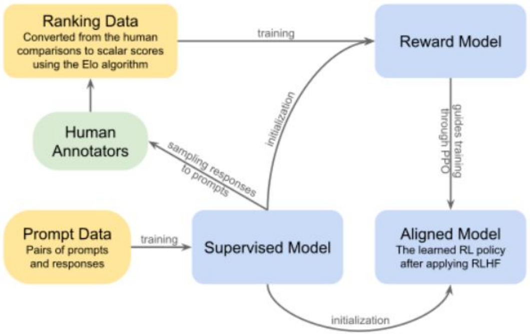


ASR

Reinforcement Learning with Deep Learning

- We will learn some basics of reinforcement learning (RL) and how they are used in modern deep learning systems:
  - Core concepts: agent, environment, states, actions, rewards
  - learning objectives in RL
  - Policy gradient methods
  - Transformers in RL (eg, Decision Transformer)
  - RL with Human Feedback → used in Chat GPT





Topics we won’t cover in the course

- Classical machine learning methods:
  - decision trees
  - SVMs
- Classical computer vision methods:
  - HOG, SIFT
  - 3D geometry
- Classical natural language processing methods:
  - bag of words
  - n-grams
  - topic modelling
- Distributed parallel training and training state-of-the-art deep learning systems
- Model quantization and efficiency

Course Logistics

Course Logistics

- Instructor and contact:
  - Office hours: Monday 09:00 am - 11:00 am
  - Office: Edificio B, Room B305
  - Book an appointment by sending an email to me
- Teaching format:
  - Lectures will cover the theoretical concepts
  - Labs for coding sessions (date of lab sessions will be announced well in advance)
  - Please bring your laptop during the labs
- Assignments:
  - Two assignments will be given (mostly coding-based assignments)
  - Should be done in a team of two students
  - You will get at most 2 weeks to finish the assignment
  - Instructions to submit assignments will be given later
  - Late policy: 2 free late days in total

Course Logistics

- Evaluation:
  - Assignments will carry 5 points each
  - Written exam will carry 30 points
  - Final score will be a weighted score and rescaled to 30
  - Example 1:
    - Assignment 1: $4/5$
    - Assignment 2: $3/5$
    - Written exam: 25/30
    - Final score: (4+3+25)/(5+5+30)*30 = 24
  - Example 2:
    - If you don't submit the assignments then the maximum you can score: 30/(5+5+30)*30 = 22.5

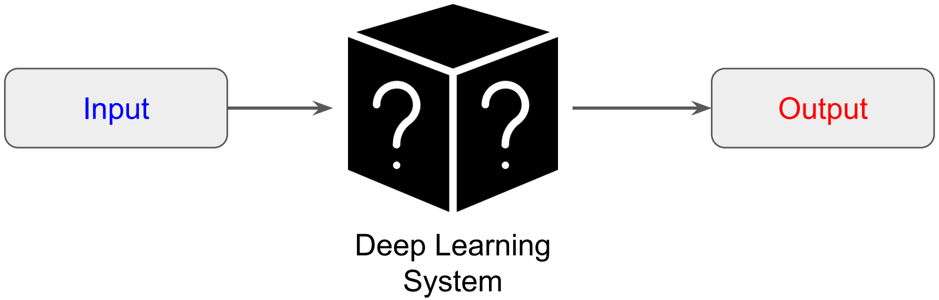
Course Logistics

- Additional resources
    - Deep Learning
        - Bit rigorous but great book
        - Link: https://www.deeplearningbook.org/
    - Dive into deep learning
        - Nice and interactive
        - Link: https://d2l.ai/index.html
    - Alice's Adventures in a Differentiable Wonderland
        - Has Italian translation too
        - Link: https://www.sscardapane.it/alice-book/
    - CS231n
        - Lecture notes for preliminary concepts
        - Link: https://cs231n.github.io/
    - I will link to other resources in the slides as the course progresses

Course Logistics

- Course homepage
  - Check here for syllabus: [link]
- Course materials
  - I will upload course materials on Moodle
  - Search for “DEEP LEARNING - 38113” in Moodle (link: https://elearning15.unibg.it/) and enroll if you aren’t already
  - Teaching materials will be made available before the lectures
- If you have any questions:
  - Feel free to send me an email
  - Please include [DeepLearning] in the email subject
  - Please except a maximum delay of 2 working days before getting a response
  - Otherwise ask me in the classroom during the breaks and after lectures

Summary



- A deep learning system is a computational model that:
  - Takes an input, produces an output based on the rules it has learned from examples
  - Can take any kind of input and output based on your design
- Deep learning has revolutionized the field of AI and has numerous applications in:
  - Computer vision
  - Natural language processing
  - Audio processing
  - Robotics
  - … and many more

LIVE Demo